# MAT2040 TUTORIAL 13

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# Question 1

Apply Gram-Schmidt orthogonalization to the following sequence of vectors in $\mathbb{R}^3$:

$$\begin{bmatrix}1\\2\\0\end{bmatrix}, \begin{bmatrix}8\\1\\-6\end{bmatrix}, \begin{bmatrix}0\\0\\1\end{bmatrix}$$

# Solution

$$x_1 = \begin{bmatrix}1\\2\\0\end{bmatrix}, x_2 = \begin{bmatrix}8\\1\\-6\end{bmatrix}, x_3 = \begin{bmatrix}0\\0\\1\end{bmatrix}.$$

$$v_1 = x_1 = \begin{bmatrix}1\\2\\0\end{bmatrix}.$$

$$v_2 = x_2 - \frac{\langle x_2, v_1\rangle}{\langle v_1, v_1\rangle} v_1 = \begin{bmatrix}8\\1\\-6\end{bmatrix} - \frac{10}{5}\begin{bmatrix}1\\2\\0\end{bmatrix} = \begin{bmatrix}6\\-3\\-6\end{bmatrix}.$$

$$v_3 = x_3 - \frac{\langle x_3, v_1\rangle}{\langle v_1, v_1\rangle} v_1 - \frac{\langle x_3, v_2\rangle}{\langle v_2, v_2\rangle} v_2 = \begin{bmatrix}0\\0\\1\end{bmatrix} - \frac{0}{5}\begin{bmatrix}1\\2\\0\end{bmatrix} - \frac{-6}{81}\begin{bmatrix}6\\-3\\-6\end{bmatrix} = \frac{1}{9}\begin{bmatrix}4\\-2\\5\end{bmatrix}.$$

$$u_1 = \frac{v_1}{\|v_1\|} = \frac{1}{\sqrt{5}}\begin{bmatrix}1\\2\\0\end{bmatrix} = \begin{bmatrix}\frac{1}{\sqrt{5}}\\\frac{2}{\sqrt{5}}\\0\end{bmatrix}.$$

$$u_2 = \frac{v_2}{\|v_2\|} = \frac{1}{9}\begin{bmatrix}6\\-3\\-6\end{bmatrix} = \begin{bmatrix}\frac{2}{3}\\-\frac{1}{3}\\-\frac{2}{3}\end{bmatrix}.$$

$$u_3 = \frac{v_3}{\|v_3\|} = \frac{1}{\sqrt{5}/3}\begin{bmatrix}\frac{4}{9}\\-\frac{2}{9}\\\frac{5}{9}\end{bmatrix} = \begin{bmatrix}\frac{4}{3\sqrt{5}}\\-\frac{2}{3\sqrt{5}}\\\frac{5}{3\sqrt{5}}\end{bmatrix}.$$

# Question 2

Assume $A$ is a $n \times n$ matrix, and $A^2 - 4A + 3I = 0$. If $\lambda$ is the eigenvalue of $A$, please show $\lambda = 1$ or $3$

# Solution

Since $\lambda$ is the eigenvalue of $A$, we have

$$Ax = \lambda x \quad (x \neq 0).$$

then, we can get

$$A^2x = \lambda Ax = \lambda^2 x \quad (x \neq 0).$$

For the equation $A^2 - 4A + 3I = 0$, we have

$$\begin{aligned}
&(A^2 - 4A + 3I)x = 0 \\
&(\lambda^2 - 4\lambda + 3)x = 0 \quad (x \neq 0) \\
&\lambda^2 - 4\lambda + 3 = 0 \\
&\lambda_1 = 3, \lambda_2 = 1.
\end{aligned}$$

# Question 3

Find the QR-factorization of $A = [a_1, a_2, a_3]$

$$\begin{bmatrix} a_1 & a_2 & a_3 \end{bmatrix} = \begin{bmatrix} -1 & -1 & 1 \\ 1 & 3 & 3 \\ -1 & -1 & 5 \\ 1 & 3 & 7 \end{bmatrix}$$

# Solution

$$\begin{bmatrix} a_1 & a_2 & a_3 \end{bmatrix} = \begin{bmatrix} -1 & -1 & 1 \\ 1 & 3 & 3 \\ -1 & -1 & 5 \\ 1 & 3 & 7 \end{bmatrix}$$

$$= \begin{bmatrix} q_1 & q_2 & q_3 \end{bmatrix} \begin{bmatrix} R_{11} & R_{12} & R_{13} \\ 0 & R_{22} & R_{23} \\ 0 & 0 & R_{33} \end{bmatrix}$$

**First column of $Q$ and $R$**

$$\tilde{q}_1 = a_1 = \begin{bmatrix} -1 \\ 1 \\ -1 \\ 1 \end{bmatrix}, \qquad R_{11} = \|\tilde{q}_1\| = 2, \qquad q_1 = \frac{1}{R_{11}}\tilde{q}_1 = \begin{bmatrix} -1/2 \\ 1/2 \\ -1/2 \\ 1/2 \end{bmatrix}$$

# Solution

**Second column of $Q$ and $R$**

- compute $R_{12} = q_1^T a_2 = 4$
- compute

$$\tilde{q}_2 = a_2 - R_{12} q_1 = \begin{bmatrix} -1 \\ 3 \\ -1 \\ 3 \end{bmatrix} - 4 \begin{bmatrix} -1/2 \\ 1/2 \\ -1/2 \\ 1/2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

- normalize to get

$$R_{22} = \|\tilde{q}_2\| = 2, \qquad q_2 = \frac{1}{R_{22}} \tilde{q}_2 = \begin{bmatrix} 1/2 \\ 1/2 \\ 1/2 \\ 1/2 \end{bmatrix}$$

# Solution

**Third column of $Q$ and $R$**

- compute $R_{13} = q_1^T a_3 = 2$ and $R_{23} = q_2^T a_3 = 8$
- compute

$$\tilde{q}_3 = a_3 - R_{13}q_1 - R_{23}q_2 = \begin{bmatrix} 1 \\ 3 \\ 5 \\ 7 \end{bmatrix} - 2\begin{bmatrix} -1/2 \\ 1/2 \\ -1/2 \\ 1/2 \end{bmatrix} - 8\begin{bmatrix} 1/2 \\ 1/2 \\ 1/2 \\ 1/2 \end{bmatrix} = \begin{bmatrix} -2 \\ -2 \\ 2 \\ 2 \end{bmatrix}$$

- normalize to get

$$R_{33} = \|\tilde{q}_3\| = 4, \qquad q_3 = \frac{1}{R_{33}}\tilde{q}_3 = \begin{bmatrix} -1/2 \\ -1/2 \\ 1/2 \\ 1/2 \end{bmatrix}$$

# Solution

**Final result**

$$\begin{bmatrix} -1 & -1 & 1 \\ 1 & 3 & 3 \\ -1 & -1 & 5 \\ 1 & 3 & 7 \end{bmatrix} = \begin{bmatrix} q_1 & q_2 & q_3 \end{bmatrix} \begin{bmatrix} R_{11} & R_{12} & R_{13} \\ 0 & R_{22} & R_{23} \\ 0 & 0 & R_{33} \end{bmatrix}$$

$$= \begin{bmatrix} -1/2 & 1/2 & -1/2 \\ 1/2 & 1/2 & -1/2 \\ -1/2 & 1/2 & 1/2 \\ 1/2 & 1/2 & 1/2 \end{bmatrix} \begin{bmatrix} 2 & 4 & 2 \\ 0 & 2 & 8 \\ 0 & 0 & 4 \end{bmatrix}$$